Technology fields, automation and task reinstatement

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Abstract

This paper develops a spatial task-based model of automation and reinstatement. Building on the geometric task-space framework of Storck and Andersson (2026), occupations are represented as bundles of tasks in a two-dimensional space of work, and the price of skill is a directional field over that space. Depth is rewarded in socio-cognitive directions but not in service and manual directions. Technologies are represented as effectiveness fields with position, reach, and intensity. Automation occurs where capital is the cheaper producer, so, within the reach of a technology, capital first takes over the tasks whose human performance is most highly priced.

Reinstatement is modeled as task creation at the boundary of a technology field. New tasks survive as human work only where labour retains a cost advantage, and they attach to existing occupations only where the priced capabilities they require are already held. The same operator therefore distinguishes three outcomes: occupational redefinition, capital capture of new tasks, and unbound tasks that mark where new occupations may emerge. Reinstatement is therefore not automatically wage-protective. It benefits workers only where new tasks bind to priced capabilities; otherwise it may restore employment while leaving wages depressed. The model therefore turns reinstatement from an aggregate labour-share mechanism into a spatial allocation problem.

The model is closed with a partial demand channel. Automation lowers unit cost where capital replaces labour, and under competitive pricing the saving is passed to consumers. Demand elasticity then determines whether a cheapened direction draws labour in or releases it. Because the cost saving is second-order at the adoption margin, displacement dominates unless demand is elastic or infra-marginal savings are large. We illustrate the framework with two technologies: a broad cognitive field calibrated to task-level AI exposure, and a narrow manual field placed to represent industrial automation. The results show how the same reinstatement mechanism can have different employment and wage effects depending on where the technology acts in task space and how the output market responds.

1 Introduction

Automation does not act on the labour market as a whole. It acts on particular tasks, within particular occupations, in particular parts of the division of labour. Workers accumulate human capital in specialized domains, occupations combine multiple tasks into bundles, and technologies vary in where they act, how far they reach, and whether capital can perform the exposed work more cheaply than labour. A model of automation therefore needs to answer not only whether new tasks are created, but where they appear, which workers can take them, and whether the tasks that return to labour carry the same priced content as the tasks that were displaced.

The task-based literature provides the canonical mechanism. Automation displaces labour from tasks that capital can perform, while reinstatement creates new tasks in which labour retains a role (Acemoglu and Restrepo 2018; Acemoglu and Restrepo 2019). In the one-dimensional framework, reinstatement raises the labour share because new tasks are placed where labour holds the comparative advantage. That is a powerful result, but it leaves an important question under-specified: which workers benefit? If labour enters as one factor, or as a small number of aggregate skill groups, reinstatement can be studied in aggregate but not located in the differentiated structure of work.

This paper introduces the concept of a technology field: a representation of a technology as an effectiveness field over a two-dimensional task space, with a position, a reach, and an amplitude. The task space is the polar geometry of Storck and Andersson (2026), in which occupations are bundles of tasks calibrated from semantic embeddings of O*NET task statements; the angular coordinate captures domain orientation and the radial coordinate specialization depth. That paper establishes the geometry and shows that the market price of skill is a field over it rather than a scalar, with depth rewarded in socio-cognitive directions and weakly or not at all in service and manual directions. The present paper is the central extension of that framework to technological change: it gives technologies a location in the same space as the work they act on, and their consequences then depend jointly on where they act and on how the price field values the work they touch. The representation separates technical exposure from economic incidence by construction, and the distinction carries the paper's results.

The central mechanism is the interaction between a technology field and the directional price of work. Capital bears a task only where it is the cheaper producer. Within the reach of a technology, the adoption margin is therefore crossed first where human work is highly priced. Automation does not simply remove occupations; it strips paid content from the bundles that define them. An occupation may remain recognizably the same while being paid for less of what it does.

Reinstatement is modeled as task creation at the boundary of the technology field. The gradient of the field identifies where automated outputs must be validated, coordinated, integrated, or translated into adjacent work. But seeded tasks do not automatically become human work. They survive as human tasks only where labour remains cheaper than capital, and they attach to an existing occupation only where that occupation already holds the priced capabilities the new task requires. The model therefore distinguishes occupational redefinition from the birth of new occupations. If the task can be absorbed by an existing bundle, the occupation is redrawn. If no occupation can bind it, the task remains unbound; where such unbound tasks concentrate, the model marks a candidate region for occupational emergence. The empirical counterpart — where new work has emerged and which workers captured it — is studied by Autor, Chin, et al. (2024); the geometry recasts it as a question of location relative to priced capabilities.

This changes the meaning of reinstatement. Task creation is not automatically worker-benefiting. Reinstatement in a low-priced manual or service direction can return work without restoring pay. The relevant question is not whether automation creates new tasks, but where those tasks appear relative to the priced capabilities workers already hold.

Employment effects also depend on output demand. Automation lowers unit cost where capital replaces labour. Under competitive pricing, the cost saving is passed to consumers, and demand elasticity determines whether the cheapened direction expands or contracts employment. Elastic demand draws labour into the automated direction; inelastic demand releases it. But the saving is second-order at the adoption margin: the marginal task that capital just takes is produced at approximately the same cost as labour. Large employment gains therefore require either elastic demand or large infra-marginal savings deep inside the technology's core. This is the spatial form of the demand mechanism emphasized by Bessen (2019) and the second-order automation margin of Acemoglu and Restrepo (2018).

We implement the framework with two illustrative technologies. The first is a broad cognitive field calibrated to task-level AI exposure and located in the high-priced socio-cognitive region. The second is a narrow manual field placed on the low-priced technical-physical arc to represent industrial automation. The comparison is deliberately asymmetric: one field is data-calibrated, the other hand-placed, because no corresponding task-level exposure surface is available for the manual case. The aim is not causal identification but mechanism: to show how the same economy responds differently when technology acts in different regions of task space.

We also confront the model with observed occupational wage changes between 2019 and 2025. A naive directional check passes: occupations whose bundles are pushed further down the price gradient exhibit weaker wage growth. We then show that the pass is uninformative. Because the price gate targets dear work, the model's predicted wage pressure is nearly collinear with the baseline wage level, and the window is dominated by the pandemic-era compression of the US wage distribution (Autor, Dube, and McGrew 2023), which makes the same cross-sectional prediction. Once baseline wage is held fixed, the mechanism adds nothing, and a vintage split shows that the entire raw correlation lies in the pre-deployment window. We report this as a finding about what the cross-section can identify, not as support. The variation the model calls for is within-occupation task recomposition over time, left to future empirical work. The exercise doubles as a caution for the growing literature that correlates AI exposure with recent wage changes.

The paper proceeds as follows. Section 2 introduces the spatial occupational geometry. Section 3 defines the price field and shows how it can be interpreted as an assignment-equilibrium object rather than a mere fitted wage surface. Section 4 develops the technology field, the adoption margin, displacement, reinstatement, and the bundle operator. Section 5 closes the model with employment sorting and the output-demand channel. Section 6 describes the empirical implementation and calibration. Section 7 presents the illustrative cases. Section 8 confronts the model with observed occupational wage changes and shows what the 2019–2025 cross-section can and cannot identify. Section 9 discusses implications and limitations.

2 The Spatial Occupational Geometry

The model builds on the task-space geometry developed in Storck and Andersson (2026) where occupations are defined from their task-bundles, with all tasks distributed on a two-dimensional polar disk. Each task has a location $\mathbf{r} = (\xi, \chi)$, where ξ is the angular domain of the work and χ is radial depth or specificity. Each occupation is a weighted bundle of tasks,

$$b_o(\mathbf{r}),$$

normalized to one before technology acts. The occupation's position is given by the centroid,

$$\mu_o = \int \mathbf{r}\, b_o(\mathbf{r})\, d\mathbf{r},$$

and thus represents the point in task space where the weighted average of the task bundle is located. The coordinates are constructed in Storck and Andersson (2026): the 17,606 O*NET task statements are embedded with a sentence encoder, projected onto their first two principal components, and expressed in polar form. The orientation is frozen to a fixed reference, so positions are comparable across runs, encoders, and database releases. That paper validates the geometry against wages, education, and occupational classifications; this paper takes it as given.

The geometry matters because technologies act on tasks rather than on occupations as indivisible units. A technology may affect part of an occupation's bundle while leaving the rest intact. The relevant unit of exposure is therefore the task location, and the relevant unit of worker consequence is the rewritten occupational bundle.

The disk has a fixed compass. The east holds human-centered work: instruction, communication, advice. The north holds analytical work: research and investigation.

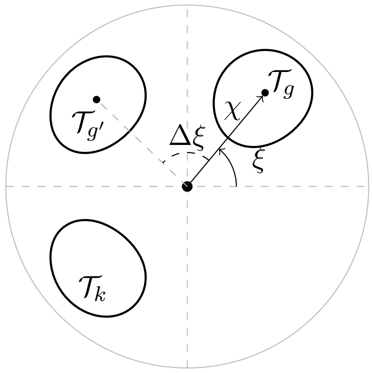

Figure 1: The polar geometry represents occupations as bundles of tasks characterised by direction, specificity and reach (Storck and Andersson 2026).

The west holds technical-physical work: mechanical and manual execution. The south holds service work: direct service and operational support. Tasks near the origin are generic activities found in many occupations, such as organizing or documenting; tasks near the rim are domain-specific. The radial coordinate χ measures this specificity, and an occupation deep in a direction is one whose tasks are both specific and aligned.

Figure 2 places nine familiar occupations on the disk. Lawyers (5°) and Pathologists (44°) sit in the interpersonal-analytical east and northeast. Software Developers (75°) and Chemical Engineers (109°) lie in the analytical north. Machinists (164°) and Carpenters (196°) occupy the technical-physical west, Dishwashers (239°) and Waiters (272°) the service south, and Biology Teachers (340°) the instructional southeast. Throughout the paper, “the socio-cognitive north” refers to the analytical-interpersonal arc from east to north, “the manual west” to the technical-physical direction, and “the service south” to the service direction.

The geometry also matters because the labour market does not price the space uniformly. The return to depth is directional. Depth in the socio-cognitive direction is rewarded, while depth in service and manual directions is weakly rewarded or unrewarded. This directional price structure is imported from Storck and Andersson (2026) and used here to study automation. It is what allows the model to distinguish between reinstatement that restores high-priced work and reinstatement that restores low-priced work.

3 The Price of Work as a Field

It was shown by Storck and Andersson (2026) that the market pays the price $\Pi(\mathbf{r})$ for a unit of work performed at location $\mathbf{r}$ in the polar task space. The directional structure enters through the price that the market pays for work content and specificity in different sectors.

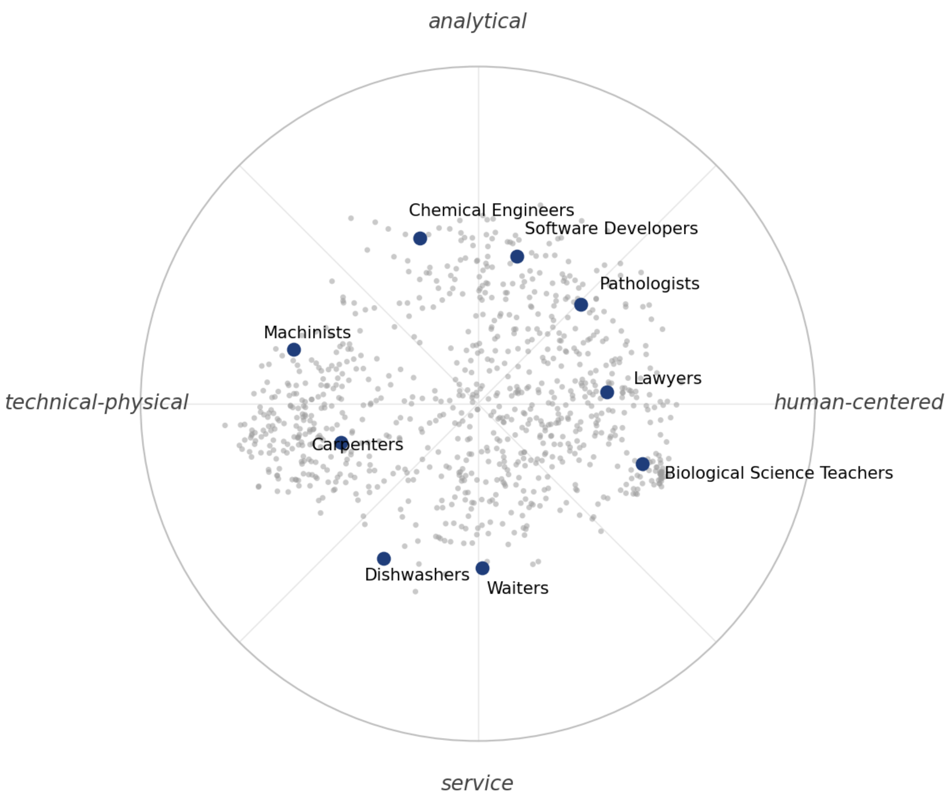

Figure 2: Occupational centroids on the task disk, with the four directional poles of Storck and Andersson (2026) and nine example occupations. Grey points are the 878 O*NET occupations; the rim is the task boundary $\chi = 1$.

3.1 The capability plane

The price field is grounded in a structural regularity of the occupational geometry. The capability content of work is, to first order, organized by a two-dimensional plane in capability space:

$$v_o = \bar{v} + \chi_o\Big(\cos\xi_o\, u_1 + \sin\xi_o\, u_2\Big) + \varepsilon_o, \tag{1}$$

where v_o is the capability profile of occupation o, $\bar{v}$ is the mean profile, and u_1, u_2 are the two poles of the plane. The disk is the polar representation of this plane: ξ is phase and χ is amplitude.

Under this structure, distance in task space is interpretable as distance in capability space, and capability requirements can be read as fields over the disk. For each capability cluster k,

$$q_k(\xi, \chi) = \bar{v}_k + \chi\Big(u_{1,k}\cos\xi + u_{2,k}\sin\xi\Big). \tag{2}$$

The clusters are the four directional families of O*NET descriptors identified in Storck and Andersson (2026): social-cognitive skills (S1), technical-physical skills (S2), social-cognitive abilities (A1), and technical-physical abilities (A2). S1 and A1 peak in the north and east, S2 in the west and northwest, A2 in the south and west. The market prices S1 strongly and S2 weakly; A1 and A2 carry no wage premium once the others are held fixed (Section 6.4).

The empirical implementation tests this postulate directly using O*NET capability descriptors that do not enter the construction of the task coordinates. The leading two components account for roughly three quarters of capability deviation variance and align strongly with the task disk. Higher-order structure is measured as texture and carried in sensitivity runs, not as part of the core theory.

3.2 The directional price field

If capability content is locally planar and capabilities are priced by a linear functional, then the log price of work is affine in the capability content of its location:

$$\ln \Pi(\mathbf{r}) = \kappa + \sum_k p_k q_k(\mathbf{r}).$$

Under the capability plane this implies a first-harmonic directional depth return. We carry the empirical reduced form as

$$\ln \Pi(\mathbf{r}) = m_0 + m_1\cos\xi + m_2\sin\xi + \chi\Big(m_3 + m_4\cos\xi + m_5\sin\xi\Big). \tag{3}$$

The depth interaction is the theoretical object: it is what makes depth valuable in some directions and not in others. The level terms capture measured structure outside the exact plane.

The coefficients are estimated once on the occupational wage cross-section and then held fixed under technology shocks. This is a deliberate partial-equilibrium choice. The technology model below asks how a given price structure mediates automation and reinstatement. It does not re-solve the full assignment of workers and tasks after the shock.

The step from occupational wages to task-location prices is testable. The field is estimated at occupational centroids, where wages are observed, but the model prices an occupation as the integral of the field over its task bundle:

$$w_o = \int \Pi(\mathbf{r})\, b_o(\mathbf{r})\, d\mathbf{r}. \tag{4}$$

Empirically, bundle-integrated wages reproduce observed wages about as well as centroid evaluation, which licenses operating the field at the task level.

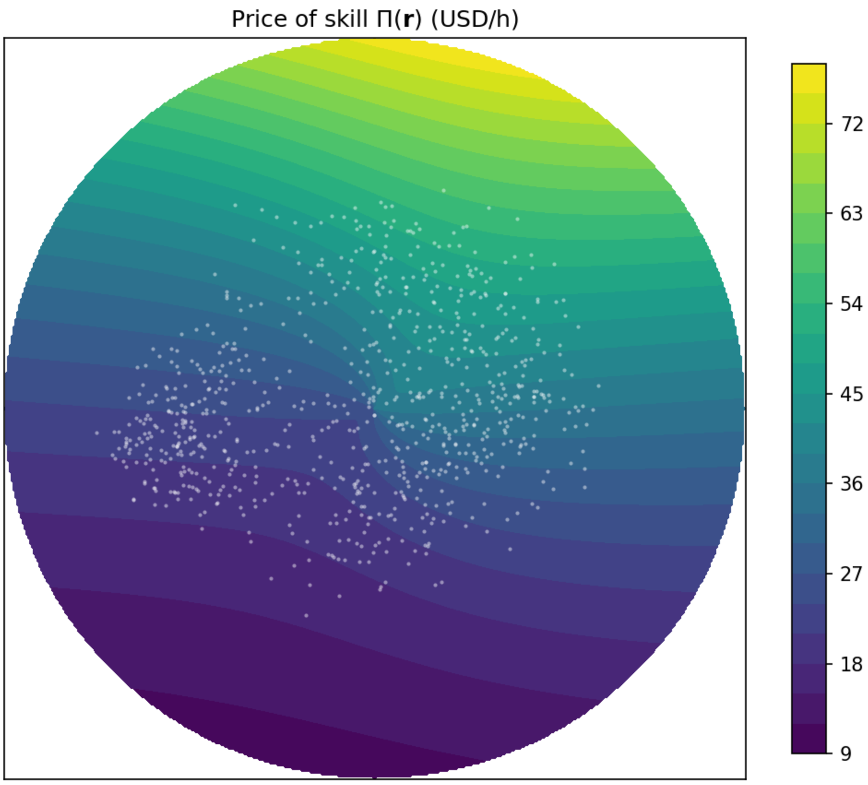

Figure 3: The price of skill $\Pi(\mathbf{r})$ over the task disk. Depth is rewarded toward the socio-cognitive north and weakly rewarded or unrewarded toward the manual west and the service south.

3.3 Assignment interpretation

The field $\Pi(\mathbf{r})$ is used as a fixed reduced form in the main model, but it is not atheoretic. It can be interpreted as the price surface generated by a multidimensional assignment

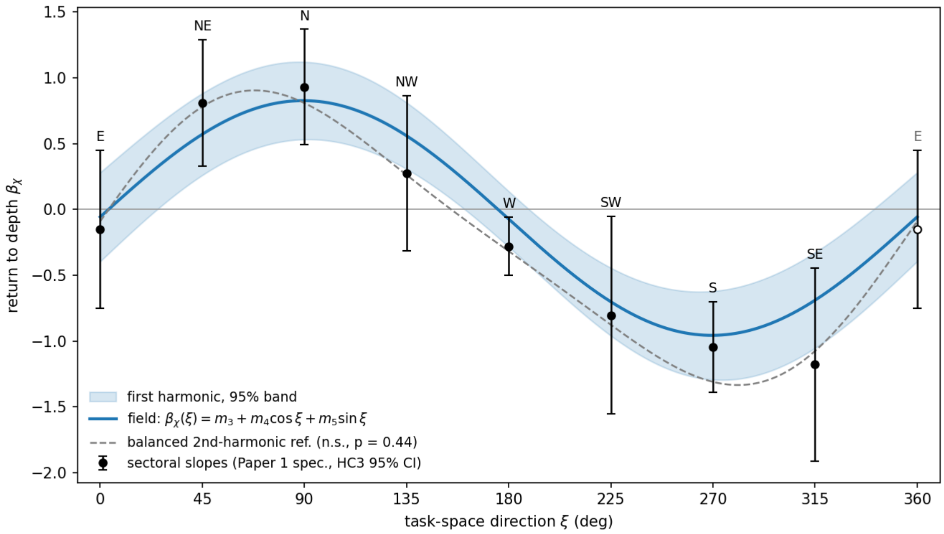

Figure 4: Directional return to depth, re-estimated on frozen inputs and compared to Storck and Andersson (2026).

equilibrium.

The field is not a simple scarcity price. If tasks were priced by a CES aggregator under uniform demand, price would be high where output is scarce; but the socio-cognitive north is both highly priced and densely staffed, so the gradient is not a scarcity phenomenon. Nor are the capability coefficients ordinary factor prices — some are near zero or negative once other dimensions are held fixed. The field is better read as an assignment price: the shadow price of matching heterogeneous capability bundles to heterogeneous task requirements under directionally biased demand.

In the numerical assignment closure, occupations are capability types, tasks require capability vectors $q(\mathbf{r})$, and workers allocate toward tasks that pay high prices relative to their readiness. With uniform demand, the assignment recovers much of the shape of the reduced-form field (rank correlation +0.92) but produces too weak a north–south gradient (+0.08 against the observed +0.39). A single cognitive demand shifter,

$$\alpha(\mathbf{r}) = \exp(\lambda q_{S1}(\mathbf{r})),$$

steepens the gradient to +0.36 while preserving the shape (rank correlation +0.97 at $\lambda = 0.2$). A production-complementarity alternative concentrates labour in the north but makes it too cheap, because output expands faster than price: the correlation between wages and the price surface turns negative. The dearness of cognitive work is therefore better understood as demand-driven assignment pressure than as productivity superiority.

The implication for this paper is limited but important. The price field can be treated as an equilibrium object: a capability-assignment cost surface with a demand-driven cognitive premium. The comparative statics below hold that surface fixed. Re-solving the assignment after a technology shock is a general-equilibrium extension and is expected to dampen, not reverse, the mechanisms studied here.

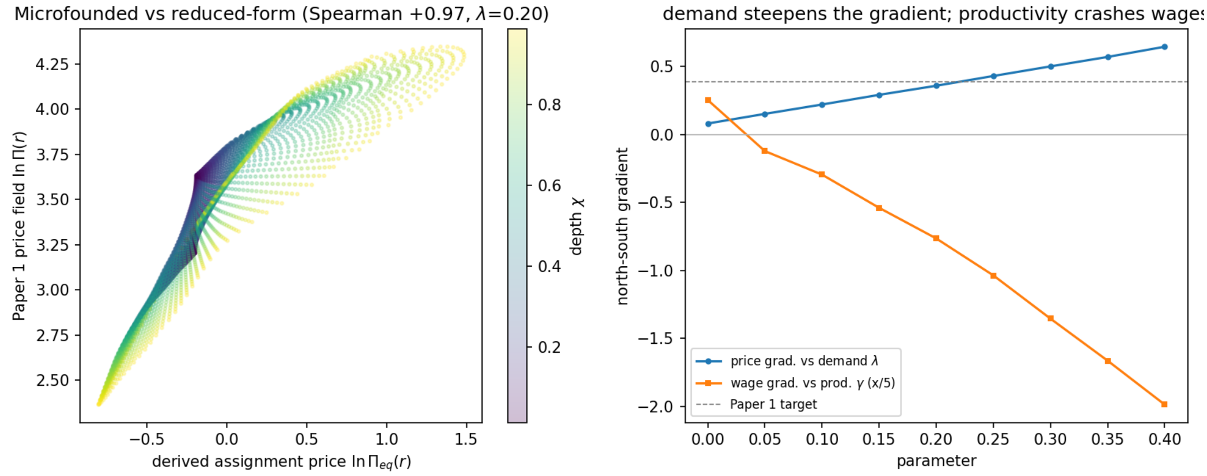

Figure 5: Assignment interpretation of the price field. The calibrated assignment price reproduces the reduced-form field closely when demand is biased toward cognitively intensive tasks.

4 Technology Fields and Task-Bundle Rewriting

This section introduces the paper's central object: the technology field, a representation of a technology over the task space of Storck and Andersson (2026). The construction is best read against the one-dimensional model it generalises. Acemoglu and Restrepo (2018) order tasks on an interval. Capital can produce every task up to a moving threshold, and each newly created task enters at the top of the index, so displacement and reinstatement both act at the two ends of a line segment. Adoption itself is already economic in that model: an automatable task goes to capital only where doing so is cheaper. Two further steps, however, rest on the shape of the line. Comparative advantage is monotone in the task index, so the newest task is by definition the task labour is relatively best at, and a ceiling on the accumulated capital ensures that this task is performed by labour (Acemoglu and Restrepo 2018). Newly created tasks that capital would win are acknowledged and left outside the analysis (Acemoglu and Restrepo 2019).

In two dimensions, each of these devices acquires empirical content. The top of the index becomes the boundary of a technology field. The monotone comparative advantage becomes the estimated field $\Pi(\mathbf{r})$, which orders work by what the market pays for it. The capital ceiling becomes a price gate applied pointwise, so whether labour keeps a newly created task is settled location by location, and the tasks capital wins are counted rather than excluded. Above all, the line gives new work a single place to appear; the disk gives it a full circle around the field. Which occupations end up holding the new work stops being fixed by construction and becomes the outcome the model must deliver. That question is what the second dimension opens, and answering it occupies the rest of the paper.

A technology is represented as an effectiveness field over task space:

$$\phi_K(\mathbf{r}) = A_K \exp\left[-\frac{1}{2}\left(\frac{\|\mathbf{r} - p_K\|}{z_K}\right)^2\right], \tag{5}$$

where p_K is the centre of the field, z_K its reach, and A_K its amplitude. A narrow field represents a specialized technology; a broad field represents a more general-purpose technology.

The field does not mean that a whole occupation is automated. At each task location, the technology can bear some micro-tasks and not others. The economic question is whether capital is the cheaper producer of those micro-tasks.

4.1 The operated share

Let $s_K \in [0, 1]$ denote the replacing character of the technology. The main simulations take $s_K = 1$. Capital performs work at $\mathbf{r}$ at effective cost

$$\frac{R}{s_K \phi_K(\mathbf{r})},$$

where R is the rental cost of capital. Labour performs the same work at price $\Pi(\mathbf{r})$. Capital is the cheaper producer where

$$s_K \phi_K(\mathbf{r}) > \frac{R}{\Pi(\mathbf{r})}. \tag{6}$$

Within the reach of the technology, this condition is crossed first where labour is highly priced. The model therefore separates technical exposure, ϕ_K, from economic incidence, $a(\mathbf{r})$.

Because a location contains a continuum of micro-tasks, the condition is smoothed into an operated share:

$$a(\mathbf{r}) = \frac{1}{1 + \exp\left(-\frac{s_K \phi_K(\mathbf{r}) - R/\Pi(\mathbf{r})}{\tau}\right)}, \tag{7}$$

where τ is the width of the adoption margin. As $\tau \to 0$, the share approaches a sharp partition; for positive τ, it is the fraction of micro-tasks at a location that capital bears.

Displacement is loss of paid human task content. For occupation o, the displaced share is

$$D_o = \int b_o(\mathbf{r})\, a(\mathbf{r})\, d\mathbf{r}. \tag{8}$$

An occupation may therefore experience substantial displacement even if no spatial region is fully automated. The margin cuts inside the bundle.

4.2 Cost saving and the second-order margin

Where capital bears the operated share, unit cost falls from $\Pi(\mathbf{r})$ to

$$c(\mathbf{r}) = (1 - a(\mathbf{r}))\Pi(\mathbf{r}) + a(\mathbf{r})\frac{R}{s_K \phi_K(\mathbf{r})}. \tag{9}$$

The realized per-unit saving is

$$\Pi(\mathbf{r}) - c(\mathbf{r}) = a(\mathbf{r}) \left[\Pi(\mathbf{r}) - \frac{R}{s_K \phi_K(\mathbf{r})} \right].$$

This saving is largest deep inside the technology's core, where capital has a strong cost advantage. It vanishes at the adoption margin, because the marginal task capital just takes is one it produces at approximately the same cost as labour. This second-order margin is central to the employment results below. Displacement begins as soon as the operated share rises, but the demand channel is weak until infra-marginal savings are large.

4.3 Reinstatement as boundary task creation

Reinstatement returns a fraction γ of displaced task mass as new tasks. The question is where these tasks appear and which occupations can bind them.

New tasks are seeded where the technology field changes most rapidly:

$$\|\nabla \phi_K(\mathbf{r})\| = \frac{1}{z_K^2} \|\mathbf{r} - p_K\| \phi_K(\mathbf{r}). \tag{10}$$

This gradient peaks at distance z_K from the technology centre. Let

$$\hat{g}(\mathbf{r}) = \frac{\|\nabla \phi_K(\mathbf{r})\|}{\int \|\nabla \phi_K(\mathbf{r})\| \, d\mathbf{r}}$$

be the normalized boundary density and let

$$M = \gamma \Delta \Gamma^D$$

be the seeded task mass, where

$$\Delta \Gamma^D = \sum_o L_o D_o.$$

The seeded density is

$$\zeta(\mathbf{r}) = M \hat{g}(\mathbf{r}).$$

A seed does not automatically become human work. It survives as human work only to the extent that labour remains the cheaper producer at its location. The surviving seed density is therefore

$$\zeta(\mathbf{r})(1 - a(\mathbf{r})),$$

while

$$\zeta(\mathbf{r})a(\mathbf{r})$$

is captured by capital.

4.4 The deficit gate

A surviving seed can attach to an existing occupation only if the occupation and the task fit each other. The fit is measured in priced capabilities. The capability deficit of occupation o at location $\mathbf{r}$ is

$$\delta_o(\mathbf{r}) = \sum_{k \in \mathcal{K}} v_k \max(q_k(\mathbf{r}) - q_{o,k}, 0), \tag{11}$$

where $q_k(\mathbf{r})$ is the requirement of capability cluster k at location $\mathbf{r}$, $q_{o,k}$ is occupation o's capability content, and v_k is the corresponding component of the price functional. In the main implementation, $\mathcal{K}$ contains the priced clusters. The corresponding surplus is $\bar{\delta}_o(\mathbf{r}) = \sum_{k \in \mathcal{K}} v_k \max(q_{o,k} - q_k(\mathbf{r}), 0)$, the priced capability the occupation holds in excess of what the task requires.

Readiness to attach the task is

$$e_o(\mathbf{r}) = \exp\Big[-\Big(\delta_o(\mathbf{r}) + \lambda_{\text{over}}\, \bar{\delta}_o(\mathbf{r})\Big)/\ell\Big]\ \exp\Big(-\|\mathbf{r} - \mu_o\|/\rho\Big), \tag{12}$$

where ℓ is a global acquisition scale, λ_{over} weights the surplus, and ρ sets the attachment locality around the occupation's centroid. The form penalizes deficit and surplus together because being able to do a task and becoming its home are different questions. The deficit records what the occupation would have to acquire; the surplus records how far the task sits below the occupation's priced level, and without the penalty an over-qualified generalist far from its core would bind seed as if it were home territory. The locality term keeps attachment near the bundle the occupation actually performs.

Setting $\lambda_{\text{over}} = 0$ and $\rho \to \infty$ gives the one-sided limit

$$e_o(\mathbf{r}) = \exp[-\delta_o(\mathbf{r})/\ell], \tag{13}$$

in which only deficits bar attachment. The limit is useful for intuition — an occupation can always absorb a task that uses less of a capability it already holds — but eq. (12) is the form the model carries. Total attachment capacity at a location is

$$C(\mathbf{r}) = \sum_o L_o e_o(\mathbf{r}).$$

The bound share of surviving seed mass is

$$\Phi(C) = \frac{C}{1 + C}.$$

Occupation o receives reinstated density

$$\iota_o(\mathbf{r}) = \zeta(\mathbf{r})(1 - a(\mathbf{r}))\Phi(C(\mathbf{r}))\frac{L_o e_o(\mathbf{r})}{C(\mathbf{r})}. \tag{14}$$

The price gate and the deficit gate act in series. The first decides whether a seed survives as human work. The second decides which occupation, if any, can bind it.

4.5 The bundle operator

The post-technology bundle is

$$b_o^{\text{post}}(\mathbf{r}) = b_o(\mathbf{r})(1 - a(\mathbf{r})) + \frac{\iota_o(\mathbf{r})}{L_o}. \tag{15}$$

The occupation is therefore not abolished. Its paid task content is redrawn. Automation strips operated content; reinstatement adds bound seeded content.

The operator does not preserve bundle mass. Its integral is

$$\int b_o^{\text{post}}(\mathbf{r})\, d\mathbf{r} = 1 - D_o + B_o/L_o,$$

where $B_o = \int \iota_o(\mathbf{r})\, d\mathbf{r}$. We do not renormalize the bundle, because doing so would assume that freed time is proportionally reallocated across remaining tasks. In this model, the mass deficit is precisely the labour released to the worker-sorting layer.

The wage change implied by bundle rewriting is

$$\Delta w_o = \int \Pi(\mathbf{r}) \left[-b_o(\mathbf{r}) a(\mathbf{r}) + \frac{\iota_o(\mathbf{r})}{L_o} \right] d\mathbf{r}. \tag{16}$$

The first term removes paid content, typically from high-priced tasks. The second term adds new human task content, but only where the deficit gate permits attachment. Reinstatement therefore carries a downward wage bias when it strips high-priced work and refills with lower-priced work.

4.6 Unbound tasks and candidate occupations

Not all surviving seed attaches. The unbound density is

$$u(\mathbf{r}) = \zeta(\mathbf{r})(1 - a(\mathbf{r}))[1 - \Phi(C(\mathbf{r}))]. \tag{17}$$

It is the density of new tasks that survive capital but that no existing occupation can perform. It is not an agent and it is not yet an occupation. It is a map of mismatch.

Each seeded location therefore has three possible fates:

$$a(\mathbf{r}) \quad \text{captured by capital,}$$

$$(1 - a(\mathbf{r}))\Phi(C(\mathbf{r})) \quad \text{bound by existing occupations,}$$

$$(1 - a(\mathbf{r}))[1 - \Phi(C(\mathbf{r}))] \quad \text{left unbound.}$$

These shares exhaust the seeded mass.

Where unbound mass is diffuse, it represents friction. Where it clusters, it marks a candidate region for occupational emergence. Whether such a region is an opportunity or merely low-paid residual work depends on the price field. Unbound mass in a high-Π region is latent skilled work awaiting a bearer; unbound mass in a low-Π region is closer to precarious task spillage.

5 Employment, Demand and Mobility

The task layer rewrites bundles. The worker layer determines how labour is reallocated after the rewrite. We close the model as a static spatial-sorting equilibrium and read technology effects as comparative statics between the pre-technology and post-technology rest points.

5.1 Task-work and output demand

The market structure is stated once and used throughout. Each location $\mathbf{r}$ hosts a local product assembled from its micro-task continuum in fixed proportions. Human task work at $\mathbf{r}$ costs $\Pi(\mathbf{r})$. Capital is supplied elastically at rental R and is paid its cost, $R/(s_K\phi_K(\mathbf{r}))$ per unit of task work; it retains no quasi-rent. The marginal cost of the local product is the blended index $c(\mathbf{r})$ of eq. (9), and competition passes cost changes into the product price.

The human part of occupation o's coverage is

$$h_o(\mathbf{r}) = b_o^{\text{post}}(\mathbf{r}),$$

and the capital part is

$$k_o(\mathbf{r}) = b_o(\mathbf{r})a(\mathbf{r}).$$

Total task-work at location $\mathbf{r}$ is

$$n(\mathbf{r}) = \sum_o L_o\Big[h_o(\mathbf{r}) + k_o(\mathbf{r})\Big] = \sum_o L_o b_o(\mathbf{r}) + \sum_o \iota_o(\mathbf{r}). \tag{18}$$

Two components of seeded mass stay outside eq. (18). Unbound tasks have no bearer and produce nothing. Capital-captured seeds are new capital-run production; pricing them requires re-solving the field and is deferred with the rest of the closure. Only bound seed enters the frozen-field economy. The fate accounting of Section 4.6 records all three.

With isoelastic demand of elasticity η around the baseline price, the quantity bought scales as

$$\left(\frac{\Pi(\mathbf{r})}{c(\mathbf{r})}\right)^{\eta},$$

and revenue scales as

$$\Pi(\mathbf{r})\left(\frac{c(\mathbf{r})}{\Pi(\mathbf{r})}\right)^{1-\eta}.$$

Define the demand multiplier

$$D(\mathbf{r}) = \left(\frac{c(\mathbf{r})}{\Pi(\mathbf{r})}\right)^{1-\eta}. \tag{19}$$

Place output is

$$V(\mathbf{r}) = D(\mathbf{r})\Pi(\mathbf{r})n(\mathbf{r})^{\beta}, \qquad \beta \in (0,1). \tag{20}$$

At $\eta = 1$, revenue is cost-invariant and $D \equiv 1$. For $\eta > 1$, cost reductions expand revenue and draw labour into the cheapened direction. For $\eta < 1$, cost reductions reduce revenue and release labour. The congestion exponent $\beta < 1$ is local capacity; its rent $(1-\beta)V$ accrues to a fixed local factor and stays outside the wage analysis.

5.2 Occupation values

The value of occupation o is

$$W_o = \beta \int h_o(\mathbf{r})D(\mathbf{r})\Pi(\mathbf{r})n(\mathbf{r})^{\beta-1}d\mathbf{r}. \tag{21}$$

Occupation values are coupled because many occupations cover the same task locations. When workers move into an occupation, they raise density where that occupation operates and reduce marginal value through congestion.

Human task work is paid its marginal value product, eq. (21). Pricing uses cost; payment uses marginal value. The two are not tied by a zero-profit condition here, because that condition re-solves the price field. The frozen-field model keeps this seam visible and defers the closure.

The paper carries three wage objects. The bundle wage $w_o = \int \Pi \, b_o$ prices a bundle at the frozen field and is used for cross-sectional levels. The occupation value W_o adds demand and congestion and is what workers sort on. The bundle wage change Δw_o of eq. (16) is the operator's gross re-pricing of the bundle and is the object taken to the data in Section 8. They agree on direction and differ in what they hold fixed.

5.3 Spatial re-sorting

Labour re-sorts from the pre-technology distribution. The mobility cost between two occupations is proportional to their distance in task space, task distance in the sense of Gathmann and Schönberg (2010):

$$d(j, o) = \|\mu_j - \mu_o\|.$$

The probability that a worker originating in occupation j moves to occupation o is

$$P_{j \to o} = \frac{\exp[(W_o - c_m \, d(j, o))/\nu]}{\sum_{o'} \exp[(W_{o'} - c_m \, d(j, o'))/\nu]}, \tag{22}$$

where c_m governs mobility cost and ν logit smoothness. Employment in occupation o is then

$$L_o = \sum_j L_j^0 P_{j \to o}. \tag{23}$$

This is a fixed point because W_o depends on $n(\mathbf{r})$, which depends on the employment vector L. Total labour is fixed; allowing aggregate labour supply to respond to total value is a later extension. The sorting layer is deliberately reduced-form: it is not a search or matching model but a static allocation rule that makes mobility costly in task-space distance and lets congestion close the equilibrium.

5.4 Labour share

Let

$$H(\mathbf{r}) = \sum_o L_o h_o(\mathbf{r})$$

be human task-work. The labour share of variable task-work income is

$$\Lambda = \frac{\int D(\mathbf{r})\Pi(\mathbf{r})H(\mathbf{r})n(\mathbf{r})^{\beta-1}d\mathbf{r}}{\int D(\mathbf{r})\Pi(\mathbf{r})n(\mathbf{r})^{\beta}d\mathbf{r}}. \tag{24}$$

The congestion exponent affects sorting but cancels from the share itself. The demand multiplier does not cancel, because it reweights the directions whose output expands or contracts. Automation lowers the share where the operated share rises; reinstatement raises it only where new human task-work binds. Λ weights human task work by the local value of task work. It records who performs the valued work; the full income statement of a location, with capital cost, congestion rent and consumer gain, belongs to the deferred closure.

6 Empirical Implementation and Calibration

This section measures the frozen objects used by the model and calibrates the illustrative technology fields. The theory objects are the capability plane and the price functional. The measurement objects are their estimated coordinates, the bundle-pricing validation, and the family wage wedge. The simulation primitives are the technology and economy parameters varied or held fixed in the case exercises.

6.1 Data and frozen inputs

All inputs are vendored from a pinned commit of the Storck and Andersson (2026) repository. The frozen inputs are the occupation and task coordinates, O*NET task relevance ratings, OEWS median hourly wages, education levels, capability cluster intensities, and raw O*NET Skills and Abilities descriptors.

Four occupation samples appear in the paper, and each use is fixed here. The coordinate universe contains 878 occupations and is the sample of every capability-side estimation and of the geometry figures. Of these, 849 occupations carry observed OEWS employment and constitute the simulation economy of Sections 5 and 7. Of these, 785 occupations carry matched median wages and form the estimation sample of the price field and the mediation regression. Finally, 725 occupations (640 distinct SOC codes) carry a median wage in both OEWS vintages and form the sample of Section 8. The attrition is data availability at each step, not selection by the model. The task universe contains 17,606 statements, of which 17,549 (99.7 percent) join the exposure data of Section 6.5 on Task ID.

6.2 Capability plane and price field

The capability plane is estimated from descriptor deviations and tested against the task disk. The leading two components account for roughly three quarters of the capability deviation variance and align strongly with the disk coordinates. The cluster fields $q_k(\xi, \chi)$ are then estimated by OLS on occupation-level capability intensities.

The price field is estimated by OLS of log median hourly wage on the polar regressors in eq. (3), on the 785 occupations of the wage sample. The estimated field, with HC3 standard errors in parentheses, is

$$\ln \Pi(\mathbf{r}) = \underset{(0.036)}{3.420} + \underset{(0.048)}{0.192} \cos\xi + \underset{(0.051)}{0.089} \sin\xi + \chi\Big(\underset{(0.090)}{-0.065} + \underset{(0.112)}{0.008} \cos\xi + \underset{(0.134)}{0.891} \sin\xi\Big), \tag{25}$$

with $R^2 = 0.523$. The isotropic depth term is indistinguishable from zero ($p = 0.47$), and the depth return is carried by a single strong first harmonic of amplitude 0.891 peaking at 89.5°, in the socio-cognitive north: depth pays only through its interaction with direction. A second-harmonic test balanced with level harmonics does not reject the single first harmonic of the depth return ($p = 0.44$); the remaining higher-order structure is mainly a level effect, not an additional depth-pricing mechanism.

Table 1: Capability plane coefficients, estimated on the full coordinate universe ($N = 878$; wages not required). Each cluster's requirement field is $q_k = \bar{v}_k + \chi(u_{1,k}\cos\xi + u_{2,k}\sin\xi)$ (eq. (2)); R^2 is the two-pole plane fit and ΔR^2 the gain from adding level harmonics and isotropic depth.

Cluster	$\bar{v}_k$	$u_{1,k}$	$u_{2,k}$	Pole	R^2	ΔR^2
S1	2.68	1.09	1.42	52°	0.74	0.001
S2	1.37	−2.22	1.23	151°	0.69	0.008
A1	3.20	0.95	1.31	54°	0.75	0.000
A2	1.57	−1.54	−0.24	189°	0.57	0.009

6.3 From centroids to bundles

The reduced-form step from occupational centroids to task locations is validated by bundle integration. The bundle-integrated wage

$$w_o = \int \Pi(\mathbf{r}) b_o(\mathbf{r}) d\mathbf{r}$$

predicts observed wages about as well as evaluating the price field at the centroid ($R^2 = 0.51$ against 0.52 at the centroid; correlation 0.72). The Jensen gap between the two is small (mean +0.018). This licenses the model's use of Π at task locations.

6.4 Gate weights and wage wedge

The gate weights v_k are the components of the price functional. They are estimated by replicating the mediation regression of Storck and Andersson (2026) on the wage sample ($N = 785$, HC3 standard errors). The gate carries the sign restriction $v_k \geq 0$: a capability enters the deficit only if the market prices it. The estimates are $v_{S1} = 0.328$ (0.065) and $v_{S2} = 0.049$ (0.021); the ability clusters carry negative or negligible estimates and are excluded by the restriction, so the priced set is $K = \{S1, S2\}$, with S1 dominant and S2 smaller.

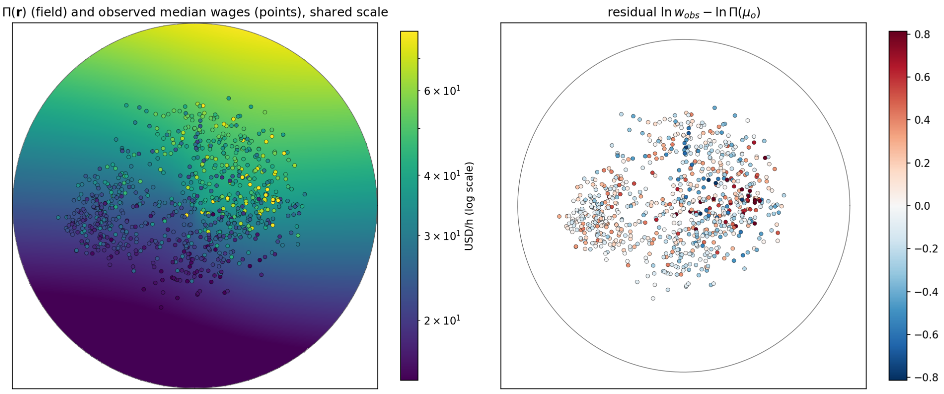

Figure 6: Bundle-integrated wage against observed wage.

The residual wage wedge is

$$\omega_o = \ln w_{\text{obs},o} - \ln w_{\text{bundle},o}.$$

The family wedge ω_g is the family-level mean of this residual. It is measured once and held fixed under technology shocks. When used, it enters the adoption gate through

$$\Pi_{\text{eff}} = e^{\omega_g}\Pi.$$

The main mechanism is reported with and without this wedge.

6.5 Technology calibration

The cognitive technology field is calibrated to task-level AI exposure: each task statement carries the exposure score $\beta_E = E1 + \frac{1}{2}E2$ of Eloundou et al. (2024). The scores are joined to the task coordinates on Task ID (17,549 of 17,606 statements, 99.7 percent) and fitted with the Gaussian field in eq. (5). The fitted field lies in the socio-cognitive north, with broad reach and moderate amplitude. The field explains a modest share of task-level exposure ($R^2 = 0.25$) but most of the smoothable between-cell exposure surface ($R^2 = 0.82$).

The modest task-level fit is a property of the rating scale, not of the field form. The exposure ratings clump at discrete values, so the between-cell share of the task-level variance — the smoothable ceiling for any field — is 0.30 on the polar grid, and the single isotropic Gaussian captures 82 percent of it. The parameters are jointly identified: the fitted centre lies inside the occupied disk ($\chi_K = 0.463$ against occupational support $\chi \leq 0.752$), so the peak amplitude is observed rather than extrapolated from a flank, and the Gauss–Newton normal matrix is well conditioned (condition number 30, with correlation -0.30 between z_K and A_K). Directional structure the isotropic field misses is carried as texture; a richer ϕ_K is an extension, not a repair.

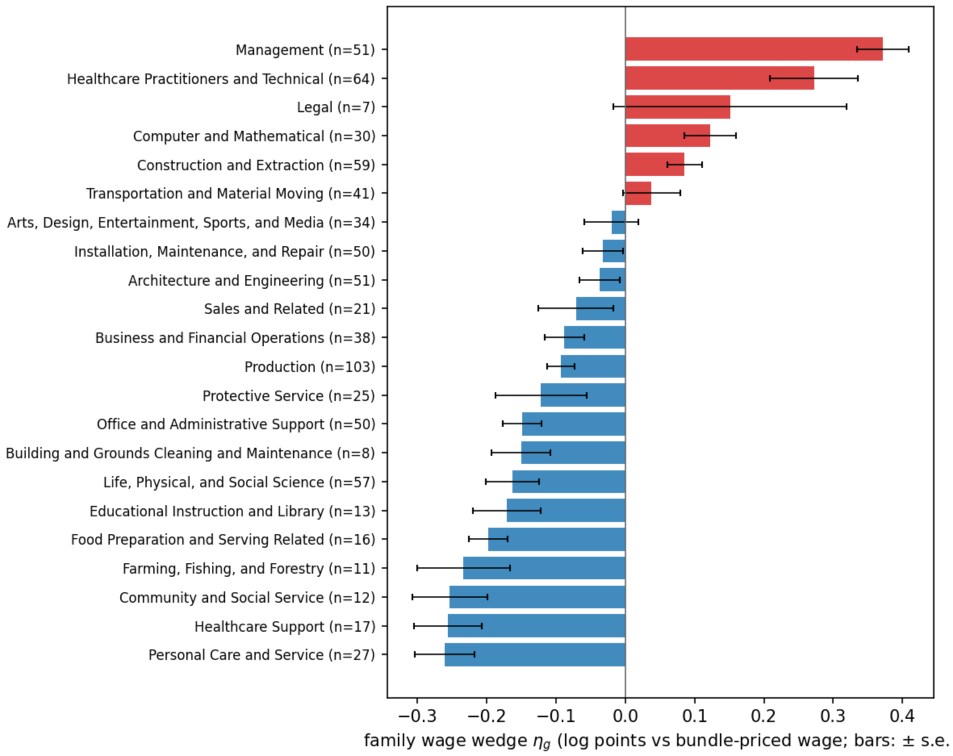

Figure 7: Family wage wedge ω_g: mean log deviation of observed wages from the bundle-priced wage, by job family. Management sits highest above the field and Personal Care and Service lowest below it; the wedge enters only the adoption margin.

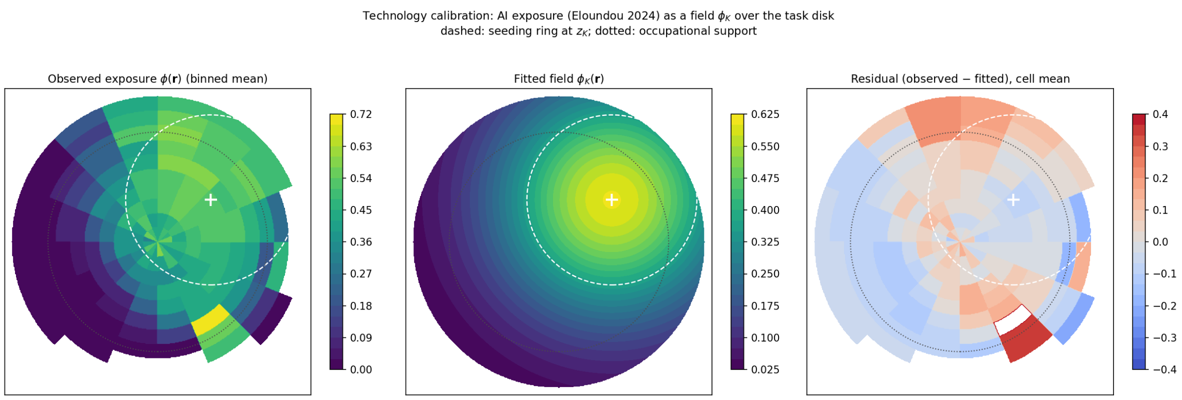

Figure 8: Task-level AI exposure surface and fitted cognitive technology field.

The manual technology field is placed by hand on the technical-physical arc. It is narrow and sufficiently intense to cross the price gate against low-priced labour. The hand placement is the controlled contrast: the same economy and the same operator, with only the field's position, reach, and amplitude changed, and no task-level exposure surface exists for industrial automation to calibrate against. The asymmetry between the two fields is disclosed, not hidden. The manual field is an illustrative industrial automation field, not an empirically calibrated exposure surface.

The economy parameters are set by explicit rules and held fixed across scenarios; Table 2 lists them. One rule is deliberately circular and is owned as such: the rental cost R is set so that the gate binds in the calibrated field's core, because the illustration requires an operative technology, and the R sweep of Section 7.1 carries the level of the results. The output-demand elasticity η is swept. Sensitivity of the headline quantities to these parameters is reported in Section 7.1.

Table 2: Economy parameters. Employment L is normalized to shares summing to one, so the attachment capacity C in $\Phi(C) = C/(1 + C)$ is of order one. Mobility values are evaluated at baseline employment under the calibrated technology.

Parameter	Value	Rule
R	18	the gate binds in the calibrated field's core: capital cost $R/(s_K\phi_K) \approx 30$ at the centre against $\Pi \approx 47$; the gate closes where $s_K\phi_K\Pi < R$
τ	0.08	width of the adoption margin, eq. (7)
β	0.5	congestion benchmark; cancels from Λ
γ	0.5	seeded fraction of displaced mass; swept below
ℓ	0.133	one within-direction SD of the priced-capability index gives readiness e^{-1}
$\rho,\ \lambda_{\text{over}}$	0.5, 1	attachment locality and over-qualification weight, eq. (12)
ν	11.6	one SD of baseline occupation value per logit unit
c_m	22.6	the median move costs one ν

Table 3: Technology calibration. The cognitive field is fitted to task-level AI exposure by least squares; the manual field is placed by hand. Fit is the R^2 on the smoothable between-cell exposure surface.

Field	ξ_K	χ_K	z_K	A_K	Identification	Fit
Cognitive	38.4°	0.463	0.583	0.603	calibrated	0.82
Manual	180°	0.45	0.30	1.20	placed	—

7 Illustrative Cases

We run two technologies through the same calibrated economy: the cognitive field and the manual field. The purpose is illustrative. The cases show what the geometry makes visible: where work is taken, where reinstatement is seeded, which occupations can bind it, how much remains unbound, and how demand elasticity changes employment consequences.

7.1 The cognitive field through the equilibrium

The cognitive field operates the disk according to the price gate. Within bands of equal technical effectiveness, the operated share orders locations by price (the within-band rank correlation $\rho(a, \Pi) = 0.99$ is a mechanical property of the takeover rule, not independent evidence). Capital therefore moves first against dear work within the exposed region.

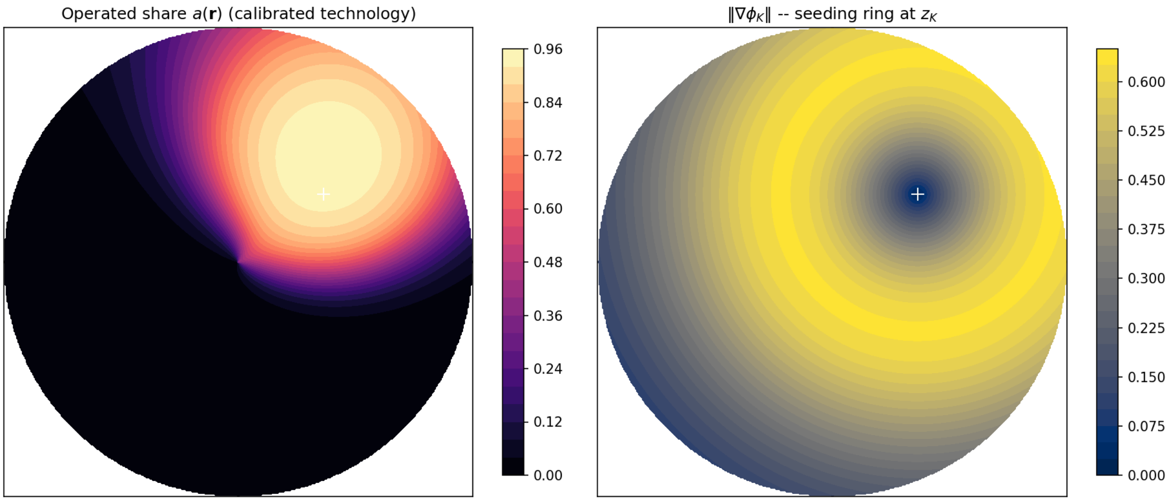

Figure 9: Operated share and reinstatement boundary for the calibrated cognitive field.

Solving the worker layer before and after bundle rewriting gives the comparative static. The labour share falls substantially under pure automation — from a pre-technology normalization of one to 0.63 — and reinstatement does not restore it, leaving it at 0.626 in the main calibration. The reason is not that task creation is absent, but that much of the seeded mass either fails to attach or attaches in lower priced regions. The geometry withholds the aggregate recovery that a one-dimensional model would assign to labour by construction. The result is robust to the family wage wedge: with the wedge in the adoption gate, the share falls to 0.647 under automation and 0.642 after reinstatement, the unbound share of seeded mass is unchanged at 68 percent, and the ordering of gaining and losing occupations is preserved.

The level of the automation drop is set by the gate. Sweeping R from 12 to 24 moves the post-automation share from 0.48 to 0.77, because R fixes how deep the technology cuts. The structure of the result is not. Across a sixteenfold range of the attachment scale, γ from 0.25 to 0.75, both margin widths, and the attachment shape, reinstatement moves the share by less than one percentage point in either direction, against an automation drop of 23 to 52 points; at the most generous corner it adds 0.4 points. The unbound share of seeded mass stays between 53 and 79 percent over the full sweep, 61 to 72 within the attachment sweep. Re-sorted mass stays at 58 percent throughout. The claim the paper rests on is the gap between task creation and wage recovery, not the level of the drop.

Re-sorting is substantial: 58 percent of employment mass relocates. Gainers are occupations whose existing capabilities let them bind the seeded tasks — engineering and mechatronics technologists. Losers include large low-priced service occupations (fast-food and retail workers, baristas, cashiers) and occupations in the technology core whose high-priced bundle content is stripped (general managers, advanced-practice nurses). The manual arc can gain employment as displaced labour re-sorts there, but that does not imply a corresponding wage recovery because depth is weakly priced in that direction.

7.2 Unbound tasks

Of the seeded mass in the cognitive case, 27 percent is captured by capital, 5 percent binds to existing occupations, and 68 percent survives capital but finds no bearer under the deficit gate. Of the seed that survives as human work, 93 percent is unbound. Its spatial distribution marks the gap between task creation and occupational absorption.

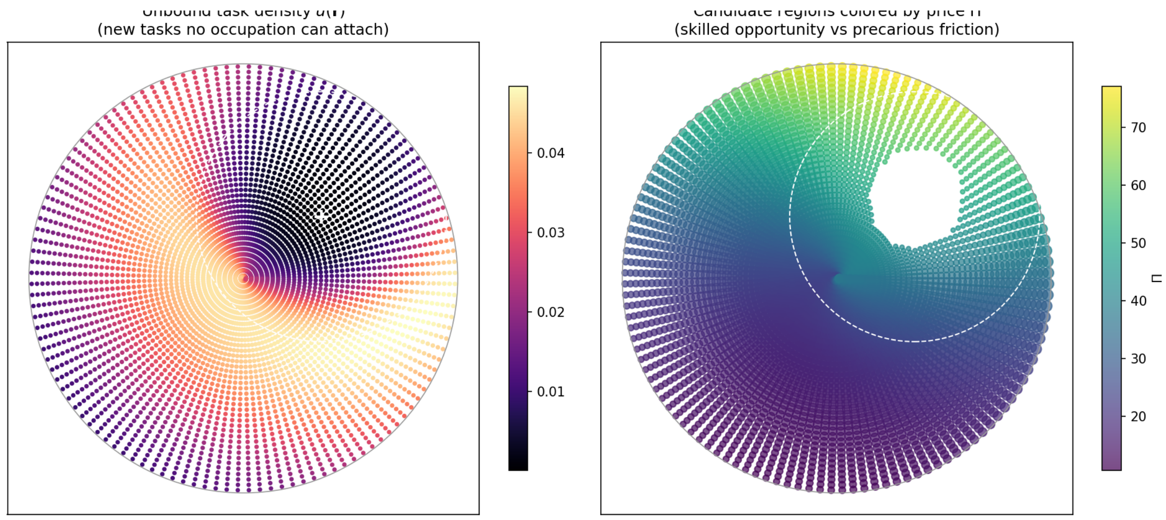

Figure 10: Unbound task density and candidate regions for occupational emergence.

The interpretation depends on price. Unbound mass in the high-priced north is latent skilled work. Unbound mass in the low-priced south is closer to low-paid friction. The model does not create new occupations; it identifies where they may become necessary.

7.3 The demand channel

The demand channel is evaluated by sweeping the output-demand elasticity η. At unit-elastic demand, cost savings leave revenue unchanged. Both technology fields release labour in their automated region — by 22 percent for the cognitive field and 6 percent for the manual one. Across plausible elasticities, displacement dominates because savings are small at the adoption margin: the price-to-cost ratio is essentially one there against roughly 1.5 deep in the technology core. Net employment growth appears only at high elasticity — the cognitive region crosses zero at $\eta \approx 7.3$ and the manual region at $\eta \approx 6.1$ — or where infra-marginal savings are large.

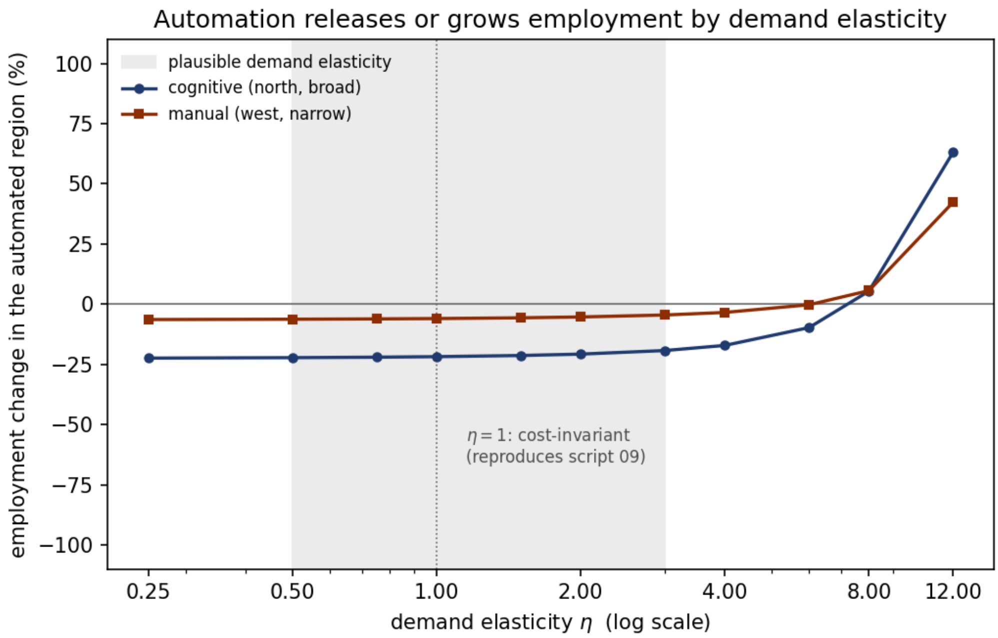

Figure 11: Employment change in each technology field's automated region as a function of output-demand elasticity.

This recovers the contrast emphasized by Bessen (2019). Automation in an inelastic-demand direction releases labour; automation in an elastic-demand direction can draw labour in. The geometry adds where this happens. Mechanised agriculture and nineteenth-century textiles occupy different places on the elasticity scale; cognitive automation is conditional on the still uncertain elasticity of demand for cognitive output.

8 What the Wage Cross-Section Can and Cannot Show

The model's empirical implication is directional. If automation and reinstatement push an occupation's bundle down the price gradient, that occupation should experience weaker wage growth than occupations whose bundles move less or remain closer to high-priced work. This section confronts that implication with observed occupational wage changes, in two steps: a naive directional check, which the model passes, and a confound analysis, which shows that the pass is uninformative in this window. We report both. The second step matters beyond this paper.

Using the calibrated cognitive field, each occupation's bundle is rewritten and its centroid shifts from μ_o to μ_o^{post}. The predicted directional wage pressure is

$$\widehat{\Delta \ln \Pi_o} = \Delta\mu_o \cdot \nabla \ln \Pi(\mu_o). \tag{26}$$

A negative value means that the bundle moves down the price gradient. Because the prediction is negative for almost every occupation, a *positive* rank correlation with wage growth means that occupations with more severe down-gradient shifts see lower wage growth.

We compare this predicted pressure to observed OEWS median hourly wage changes between May 2019 and May 2025 (the latest OEWS release), matched by SOC code; 725 of 849 occupations (640 distinct SOC codes) carry a median wage in both vintages. Because the match is many-to-one and the wages nominal, we compare ranks, which differences out the common nominal drift (mean $\Delta \ln w \approx +0.22$). The field pushes almost every bundle down-gradient (96 percent), and the naive check passes: the rank correlation between predicted pressure and observed wage change is +0.34 across occupations ($p < 10^{-20}$) and +0.35 at the SOC level; the model's own bundle wage change tracks observed changes slightly better (+0.42), and the result is stable when May 2024 is used as the endpoint (+0.39).

8.1 The confound

This looks like support for the mechanism. It is not. The predicted pressure is nearly collinear with the baseline wage level: because the price gate targets dear work, occupations with high 2019 wages are pushed furthest down-gradient (rank correlation −0.57 between predicted pressure and baseline wage). And the window contains the pandemic-era compression of the US wage distribution, in which low-wage occupations saw exceptional relative wage growth for reasons unrelated to automation (Autor, Dube, and McGrew 2023): baseline wage alone predicts wage growth at rank correlation −0.58. The model and the compression therefore make the same cross-sectional prediction, and the raw correlation cannot distinguish between them.

The data resolve the collinearity in favour of the confound. Controlling for the baseline-wage rank, the partial rank correlation between predicted pressure and wage growth is +0.02 ($p = 0.57$); the bundle wage change performs no better (+0.03). Adding the mechanism raises the rank-R^2 already achieved by baseline wage alone by 0.0003. The fine structure points the same way. The wage-growth gradient is steepest among the occupations the technology field barely reaches (rank correlation −0.64 in the least exposed tercile of ϕ_K) and nearly flat where the field is strongest (−0.14 in the most exposed tercile); in a rank regression the wage–exposure interaction is positive and significant, so exposure attenuates the gradient rather than creating it. Among above-median-wage occupations — where the mechanism strips the most and the compression matters least — the raw correlation vanishes entirely (−0.03). This is the footprint of compression at the bottom of the distribution, not of high-priced content being stripped in the exposed region.

A window split settles the matter. The OEWS 2023 vintage divides the period into 2019–2023, which contains the peak of the compression and essentially predates the deployment of the technology the field is calibrated to, and 2023–2025, the period of actual diffusion. Table 4 reports the result: the entire raw correlation lives in the pre-deployment window, and in the deployment window it is absent, along with most of the compression gradient itself.

Table 4: Window split of the directional check. Spearman rank correlations between predicted directional wage pressure and observed wage change; the partial correlation controls for the rank of the wage level at the start of each window. The baseline gradient is the rank correlation between the starting wage level and wage growth ($N = 725$).

Window	Raw	Partial | baseline wage	Baseline gradient
2019–2023 (pre-deployment)	+0.34	+0.03	−0.56
2023–2025 (deployment)	+0.04	−0.03	−0.11
2019–2025 (full)	+0.34	+0.02	−0.58

8.2 What follows

Two conclusions follow. First, the 2019–2025 occupational cross-section cannot test the mechanism. Its observable prediction and the dominant wage shock of the period load on the same index — the baseline wage — and once that index is held fixed there is no identifying variation left. The identifying variation the model calls for is within-occupation task recomposition over time: whether the paid content of exposed bundles is being redrawn in the direction the operator predicts. That test requires within-occupation microdata and is left to future empirical work. Second, the null in the deployment window is not evidence against the model. The model does not predict that two years of diffusion should be visible in occupational median wages. A strong pass in this window would itself have been suspect.

The exercise carries a wider caution. A naive correlation between AI exposure and 2019–2025 occupational wage growth has the right sign, a small p-value, and nothing to

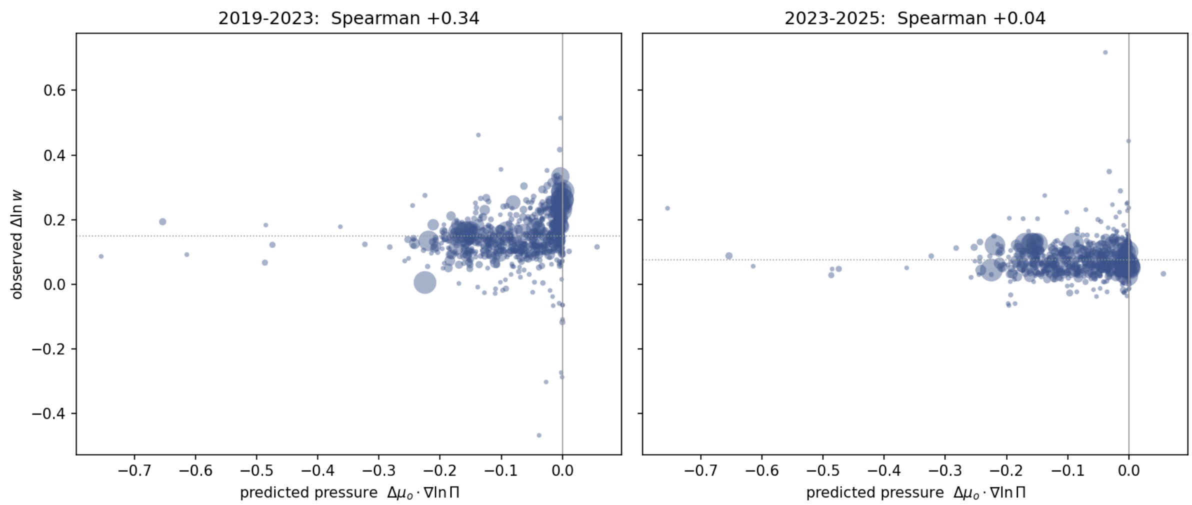

Figure 12: Predicted directional wage pressure against observed occupational wage change, split at the OEWS 2023 vintage. The raw correlation is confined to the 2019–2023 window, which contains the peak of the pandemic wage compression and predates the deployment of the technology the cognitive field is calibrated to; in the 2023–2025 window it is absent.

do with AI: it measures the pandemic compression. This is the exposure-versus-incidence distinction of Section 4.1 in empirical form. Technical exposure correlates with wage levels, wage levels drove relative wage growth in this period, and the chain of correlations manufactures apparent support for any mechanism whose cross-sectional footprint is a wage gradient.

9 Discussion

The central contribution of the paper is to locate automation and reinstatement in a priced task space. Reinstatement is not an aggregate return of labour to production. It is a spatial process in which new tasks must survive capital and then bind to existing capabilities. The same amount of task creation can therefore have different consequences depending on where it appears.

This yields a distributional result that one-dimensional models cannot state. Reinstatement is good for employment wherever it lands, because it restores human task mass. It is good for wages only where the price field rewards the work it restores. New tasks in socio-cognitive directions can restore both work and pay. New tasks in manual or service directions may restore work without restoring pay.

The model also clarifies the Turing-trap logic (Brynjolfsson 2022). Default automation strips high-priced task content from occupational bundles. Default reinstatement often refills them with tasks that are easier to bind because they demand no new priced capabilities. The occupation remains, but its paid content declines. The occupation does not vanish; its bundle is redrawn.

The demand channel is distinct from the wage channel. Automation lowers unit cost where capital replaces labour, and the output market decides whether the saving expands or contracts employment. Elastic demand can turn labour-saving technology into employment growth; inelastic demand turns it into labour release. But the adoption margin limits this channel, because savings vanish where capital just breaks even with labour. The demand channel is therefore weak exactly where automation begins.

The model keeps exposure and incidence separate. Technical exposure is what a technology can do; economic incidence is what it is paid to do. A weak raw correlation between exposure and wages does not refute the mechanism, because adoption depends on exposure relative to price. The price field turns exposure into incidence.

Several limitations bound the results, and each is a stated primitive rather than an oversight. The comparative statics hold the readiness fields e_o fixed between the two rest points: capability acquisition is treated as slower than the shock, a timescale separation in the spirit of Adão, Beraja, and Pandalai-Nayar (2024). The attachment capacity $C(\mathbf{r})$ is linear in employment. The output market is partial: each place faces independent isoelastic demand, with no cross-place substitution or income effects. The price field is held fixed under the technology shock, even though it can be interpreted as an assignment-equilibrium object. Re-solving the assignment after the shock is a first-order extension that should dampen displacement by allowing the price of crowded or cheapened work to adjust; the present paper deliberately holds $\Pi(\mathbf{r})$ fixed to isolate the spatial automation and reinstatement mechanisms.

The model is also static: it compares two rest points, and unbound tasks are read off as a density rather than accumulated as a stock. The fate accounting suggests that this is the most promising continuation of the framework. If unbound mass accumulates while occupations absorb new work at a finite pace, the speed of the shock relative to the speed of absorption becomes an economic variable in its own right, and the destination of reinstated work — which occupations end up holding it — may depend on the path and not only on the endpoint. We regard a dynamic treatment built on these observations as interesting future work.

The wage wedge also opens a connection to rent-based accounts of automation. If the residual wedge is interpreted as an occupation- or family-level rent, the same price gate implies that high-wedge work is more likely to cross the adoption margin. The within-group compression implications of that extension require rent dispersion within exposed groups and are left to separate work.

The main implication is simple. The question is not whether automation creates new tasks. The question is where those tasks appear, whether labour keeps them, and whether the workers who can bind them already hold the priced capabilities they require.

A Notation

Symbols are collected below, grouped by role.

Symbol	Meaning
Task space and the price field	
$\mathbf{r} = (\xi, \chi)$	point in task space; ξ angular, χ radial coordinate ($\chi = 0$ centre, rim $\chi = 1$)
$\mu_o = \int \mathbf{r}\, b_o(\mathbf{r})\, d\mathbf{r}$	centroid of occupation o's task bundle b_o
$\Delta\mu_o$	shift of the centroid under the technology shock
$d(o, o') = \lVert \mu_o - \mu_{o'} \rVert$	task-space distance between occupations
$\Pi(\mathbf{r})$	price field: price of capability content at $\mathbf{r}$ (reduced form, estimated from wages)
$\nabla \ln \Pi$	gradient of the log price field
$\Pi_{\text{eff}} = e^{\omega_g} \Pi$	rent-adjusted effective price
$\widehat{\Delta \ln \Pi}_o = \Delta\mu_o \cdot \nabla \ln \Pi$	predicted directional wage pressure: first-order log-price change the shift implies (Section 8)
Capabilities	
$\mathcal{K}$, k	set of capability clusters and a generic cluster
$q_k(\mathbf{r})$, $q_{o,k}$	level of capability k at $\mathbf{r}$; occupation o's content
v_k	weight of capability k
p_k, κ	per-capability price and intercept in $\ln \Pi = \kappa + \sum_k p_k q_k$
Technology field and takeover	
K	a technology field (cognitive or manual)
z_K, A_K	reach and amplitude of the field (calibration)
s_K, $\phi_K(\mathbf{r})$	replacing share of the field's effectiveness ($s_K = 1$ all replacing, 0 augmenting); ϕ_K technical exposure at $\mathbf{r}$
$a(\mathbf{r})$	operated (takeover) share at $\mathbf{r}$
R, τ	automation cost threshold (takeover compares Π to $R/(s_K\phi_K)$) and margin width
D_o	displaced share of occupation o
$\iota_o(\mathbf{r})$, $B_o = \int \iota_o\, d\mathbf{r}$	reinstatement inflow and reinstated mass
γ	reinstatement fraction ($= 0.5$)
β	congestion exponent in $V(\mathbf{r}) = D\, \Pi\, n^{\beta}$; human task-work is paid its marginal value product out of βV, capital is paid its cost, $(1-\beta)V$ is the congestion rent to the fixed factor

continued on next page

Symbol	Meaning
$\Phi(C) = C/(1+C)$	bound fraction at congestion C
Δw_o	bundle wage change of occupation o (eq. (16))
Wages and rents	
$w_{\text{obs},o}$, $w_{\text{bundle},o}$	observed wage and model bundle wage of occupation o
$\omega_o = \ln w_{\text{obs},o} - \ln w_{\text{bundle},o}$	occupation rent (residual of log wage on priced content)
ω_g	family wage wedge: the family mean of ω_o
η	demand elasticity (Section 7.3)
Equilibrium and mobility	
ℓ	acquisition scale in the readiness e_o (eq. (12))
L_0, L	baseline and re-solved employment allocation
λ	price-fit elasticity in the microfoundation
c_m, ν	mobility cost and logit smoothness in the re-sorting map
ρ, λ_{over}	attachment locality scale and over-qualification weight (eq. (12))

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